

Module 5: Assignment

Information security and assurance approaches

MCY 630 Security Architecture

Assignment Submission Instructions

Submission requirements: assignment

File format: MSWord or pdf (preferred)

Report requirements

1. Include your name and assignment title at the top of the document.
2. Number the questions and answers and answer each question in your submission in sequence.
3. For each response, include the required text as well as any necessary images and diagrams.
4. Use black font color for texts
5. If uploading an MSWord format file, view the uploaded file from Canvas to make sure that the layout of images and texts are not overlapping in the report. If the contents seem to have moved around, upload a pdf instead of an MSWord file.

Questions

(50 points)

1. (20 points) Compare and contrast each of the access control methods: MAC, DAC, RBAC, and ABAC. Present your answer in a tabular format as shown below.

Aspects	MAC	DAC	RBAC	ABAC
Control Authority				
Basis of Decision				
Flexibility				
Granularity				
Ease of Management				
User Involvement				
Scalability				
Common Use Cases				

2. (10 points) Explain the exact steps, in order, where asymmetric encryption and hashing can be used together to ensure confidentiality, integrity, and non-repudiation, when a sender *S*, and receiver *R*, exchanges a message *M*.

Provide your answer in two paragraphs. The first paragraph should specify the actions (*step 1, 2, ...*) performed by the sender *S*. The second paragraph should specify the actions (*steps 1, 2, ...*) performed by the receiver *R*, clearly stating which security property (*confidentiality, integrity, or non-repudiation*) is being ensured in which step.

3. (10 points) Compare and contrast the benefits and pitfalls of blacklisting and whitelisting for firewall security policies.
4. (10 points) Describe how an effective Data Loss Prevention (DLP) plan should function with respect to applicable strategies and practices.